ENTRY 028 Two Decoherence Bugs, One Wrong Channel — August 13, 2026 — Status: COMPLETE

Entry 028 — Dephasing Was Never the Channel That Mattered

Isolating Gate Error, Decoherence, and Readout Separately Against Aer Exposes Two Bugs in the Decoherence Term Built in Entry 023

1. Why This Investigation Started

Entry 027 left two circuits badly wrong: bell_mid_77_100 (19-point gap) and bell_far_0_126 (44-point gap), both unchanged by that entry's gate-error simplification. Rather than guess, each contributing term -- gate error, decoherence, readout -- was isolated separately in Aer against bell_mid_77_100's real 5-leg route and compared to what the emulator predicts for that term alone.

term	emulator predicts	Aer isolated	gap
gate error (depolarizing)	90.16%	91.04%	0.88 pts
readout error	92.50%	92.66%	0.16 pts
decoherence	77.6%	94.80%	17.2 pts

Gate error and readout both check out almost exactly. Decoherence is off by 17 points on its own, on a single circuit -- clearly the source of the two entries' unexplained gap.

2. Bug One: Charging Both Ends of Every SWAP

exact_dwell_cost() (Entry 024) charged decoherence to BOTH physical qubits of every SWAP leg. But only one of the two is actually carrying a logical qubit's payload at any given hop -- the other is an idle ancilla being swapped into position. v4's own BFS-guess model (route_cost(), Entry 023) already charges only the single carrier qubit per hop, matching its own stated design intent ("integrate decoherence along the CARRIER path only") -- the exact-route function introduced in Entry 024 simply didn't carry that rule over. Fixed by tracking, at each SWAP, which of its two physical qubits currently holds a pending logical qubit and charging only that one.

3. Bug Two: The Wrong Coherence Channel Entirely

Fixing bug one alone wasn't enough. The deeper problem is Entry 023's original physical reasoning: it charged decoherence using T2 (dephasing time), on the argument that Bell/GHZ states carry their information in phase. That reasoning has it backwards. Pure T2 dephasing destroys OFF-DIAGONAL coherence in the computational basis -- but every circuit in this project measures success as population landing in {00...0, 11...1}, which are diagonal elements. Verified directly: an isolated two-qubit Aer test with T1 set near-infinite and T2 = 20 μs (aggressive dephasing, no relaxation at all) gave **100.00%** Bell-pair success. Pure dephasing was completely invisible to this measurement. The same test with T1 = 20 μs instead (T2 = 2×T1, i.e. no dephasing beyond the T1 limit) gave 97.6% -- consistent with the real order-of-magnitude loss. It is T1 relaxation, population leaking toward |0⟩, that actually corrupts these outcomes by leaking probability into {01,10} -- not T2.

dephasing() now uses T1 instead of T2, wherever it's called from (both route_cost() and exact_dwell_cost()). The function name is left unchanged since every call site still refers to it, but the docstring now records this correction and the isolated test that motivated it.

4. Re-Validation Against the Entry 022 Kyiv Reference

circuit	Aer ref	gap before (Entry 027)	gap after (Entry 028)
bell_adjacent_77_78	96.07%	0.09	0.15
bell_near_77_82	48.74%	2.68	10.81
bell_mid_77_100	83.69%	19.11	3.39
bell_far_0_126	69.90%	43.79	11.21
ghz_local_77_78_79	88.53%	0.76	0.10

circuit	Aer ref	gap before (Entry 027)	gap after (Entry 028)
MEAN		13.28	5.13
MEDIAN		2.68	3.39
WORST		43.79	11.21

Entry 028 — Conclusion

Fixing both bugs together drops the mean prediction gap across the Kyiv reference set from 13.28 to 5.13 points and the worst case from 43.79 to 11.21 -- the largest single accuracy gain of the project since Entry 024's routing fix. bell_near_77_82 is the one circuit that got nominally worse (2.68 → 10.81 points), and that's worth being honest about rather than hiding: its old "good" number came from two bugs partially cancelling -- overcharging decoherence via the wrong channel on a route that happens to dwell on catastrophically low-T2 qubit 80 -- not from the model being more correct. A model that is right for the wrong reasons on one circuit and wrong elsewhere is worse than one that is consistently closer everywhere.

5. What's Still Unresolved

bell_near_77_82 and bell_far_0_126 both still carry double-digit gaps (10.81 and 11.21 points). Both routes cross qubit 80 or long chains with several low-to-mid T2 qubits, so a residual decoherence-modeling gap likely remains -- possibly in how T1 combines with the readout and gate-error terms at the point of measurement, or in whether a single T1-based formula is adequate for qubits with unusually short T1 relative to the gate durations they experience. Not chased further in this entry; flagged for Entry 029.

6. Next Step — Entry 029

Investigate the remaining bell_near_77_82 / bell_far_0_126 gap specifically on qubit 80's role (T2 = 8.51 µs, the same qubit Entry 023 originally flagged) -- isolate its T1 in Aer the same way this entry isolated dephasing generally, and check whether the single T1-based exponential formula still holds at its extreme. If it does not extend the current model's accuracy further, task #21 (extending the D-screen protocol to Sherbrooke and Cairo) is the next candidate.

Research Log — Index Addendum (Entry 028)

Entry	Title	Date	Status
Entry 028	Fixing Two Decoherence Bugs (Double-Charge, Wrong Channel)	Aug 13, 2026	Complete

emulator_v4.dephasing() now uses T1 (relaxation) instead of T2 (dephasing) -- pure T2 dephasing is invisible to a computational-basis {00,11}/{000,111} measurement. exact_dwell_cost() now charges decoherence to a route's single carrier qubit per SWAP leg, matching v4's own BFS-model design intent.